

Information Security (CS3002)

Sessional-I Exam

Date: Sep 20th 2025

Course Instructors

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Total Time (Hrs): 1

Total Marks: 42

Total Questions: 4

Student Roll No

Section

Student Signature

Instructions:

1. Attempt all questions on the answer booklet.
2. If you think some information is missing then make an assumption and state it clearly.

[This paper is NOT for BCS-7D, BCS-7H and BCS-7J sections]

CLO-1	Explain key concepts of information security such as design principles, cryptography, risk management.
CLO-2	Discuss legal, ethical, and professional issues in information security.
CLO-3	Analyze real world scenarios, model them using security measures, and apply various security and risk management tools for achieving information security and privacy.
CLO-4	Identify appropriate techniques to tackle and solve problems of real life in the discipline of information security.
CLO-5	Describe issues related to ethics in the field of information security.

CLO 1

Q1: Multiple choice questions

[7 marks]

1. A message is first encrypted with a Caesar cipher (with key 3) and then the resulting text is again encrypted with monoalphabetic substitution cipher where A is mapped to F, B to K, and so on. A cryptanalyst knows this two-step process. What is the most effective way to break this combined cipher without trying to guess the original plain text?
 - a) Brute force the Caesar cipher, then perform a frequency analysis on the intermediate text.
 - b) Perform frequency analysis on the final ciphertext, then brute force the Caesar shifts.
 - c) Brute force the monoalphabetic cipher, then brute force the Caesar cipher.
 - d) Treat the combined cipher as a single monoalphabetic substitution cipher.
2. A cryptanalyst intercepts a ciphertext that is 300 characters long. They look for repeated strings, and find that the sequence UQE repeats at positions 25 and 100, while FPR repeats at positions 50 and 150. Which of the following is the most likely key length for the Vigenère cipher?
 - a) 5
 - b) 10
 - c) 15
 - d) 25

3. What is the biggest disadvantage of symmetric encryption?
 - a) It requires more complex and time-consuming calculations.
 - b) The secret key must be transmitted securely between parties.
 - c) It uses a less secure encryption function.
 - d) It is no longer in use.
4. Which statement best describes the role of a key in encryption?
 - a) The key determines which encryption function is used, making decryption difficult if the function is unknown.
 - b) The key contains a secret encryption function along with its parameters. A password is required to activate the key.
 - c) The key only contains secret parameters used for both encryption and decryption.
 - d) The key allows users to avoid reinstalling encryption software when the technology or encryption functions change.
5. In a hybrid cryptosystem (Digital Envelope), what is the primary role of asymmetric cryptography?
 - a) To securely encrypt and exchange a symmetric session key
 - b) To encrypt the entire bulk message for speed
 - c) To generate the hash of the message for integrity
 - d) To create a digital signature for non-repudiation
6. What is the correct sequence for creating a digital signature?
 - a) Encrypt message with private key -> Hash the message
 - b) Hash the message -> Encrypt hash with recipient's public key
 - c) Hash the message -> Encrypt hash with sender's private key
 - d) Encrypt message with public key -> Hash the ciphertext
7. Why are large prime numbers (e.g., 1024 bits) used in RSA?
 - a) To make the encryption process faster
 - b) To ensure the output ciphertext is smaller than the plaintext
 - c) To make factoring the product $n = p \times q$ computationally infeasible
 - d) To increase the likelihood that e and $\phi(n)$ are co-prime

CLO 1

Q2: Short answer questions

[4 + 4 + 2 + 2 marks]

- A. Which information security property is being most directly impacted in the following scenarios? Choose one from Confidentiality, Integrity, Availability, Authenticity, Accountability
- i. During a ransomware attack, the company loses access to its systems until ransom is paid.
 - ii. An employee receives an email that looks like it is from their boss, but it was actually sent by a pretending attacker.
 - iii. University's online portal crashes on the day grades are released, leaving students unable to access results.
 - iv. An angry employee modifies payroll data so that some staff salaries are reduced.

- B. Review the below scenario, and then discuss how you could apply the following security design principles to improve the system's security. Give specific examples relevant to this system.
- least privilege
 - fail-safe defaults

Scenario: A healthcare startup has developed an online medical consultation portal. Patients register and book virtual appointments with doctors. During the consultation, doctors access the patient's medical history stored in the system. After the session, doctors update the record with prescriptions and treatment notes. Patients can later download their prescriptions or share them with pharmacies.

- C. Referring to the block cipher modes of operation, mention one benefit and one drawback of the Cipher Block Chaining (CBC) mode.
- D. Eve intercepts a message from Alice to Bob that is protected with an HMAC. She tries to modify the message as well as the MAC. Explain why will her attack fail when Bob verifies the message?

CLO 3

Q3: Practical problems

[4 + 4 + 2 + 4 marks]

- A. An adversary intercepts an encrypted message. He knows the message was first protected with a Caesar cipher with shift 3, but the output was then permuted using a rail fence cipher with a depth of 4. The final ciphertext is VWFHLBDKFUEHHXUV. Find the original plaintext message. Show your working.
- B. Consider a Vigenère cipher where sixteen hexadecimal characters (0 to F) are used as the alphabet (instead of English letters).
- For plaintext 3AE60A3 and keyword 17E, what will be the ciphertext?
 - Using your answer in (i) argue that a polyalphabetic cipher such as the above is stronger against letter frequency analysis when compared to a monoalphabetic ciphers like Caesar.
- C. Recall that DES Expansion P-box operates on a 32-bit input. Suppose we are working with a smaller version of DES in which expansion is performed on a 16-bit input. Expand the following input block, and provide the corresponding output in binary.

0010 1100 0011 0100

Expansion P-box

16	1	2	3	4	5
4	5	6	7	8	9
8	9	10	11	12	13
12	13	14	15	16	1

- D. The AES SubBytes substitution table is provided below. Given the following state matrix (in hex) for an AES encryption round, perform the SubBytes operation *followed by* the ShiftRows operation. Show the intermediate output after SubBytes, and final matrix after ShiftRows.

1B	22	34	56
4A	FF	33	E3
2B	CD	4C	5B
22	AA	CC	BB

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
0	63	7C	77	7B	F2	6B	6F	C5	30	01	67	2B	FE	D7	AB	76
1	CA	82	C9	7D	FA	59	47	F0	AD	D4	A2	AF	9C	A4	72	C0
2	B7	FD	93	26	36	3F	F7	CC	34	A5	E5	F1	71	D8	31	15
3	04	C7	23	C3	18	96	05	9A	07	12	80	E2	EB	27	B2	75
4	09	83	2C	1A	1B	6E	5A	A0	52	3B	D6	B3	29	E3	2F	84
5	53	D1	00	ED	20	FC	B1	5B	6A	CB	BE	39	4A	4C	58	CF
6	D0	EF	AA	FB	43	4D	33	85	45	F9	02	7F	50	3C	9F	A8
7	51	A3	40	8F	92	9D	38	F5	BC	B6	DA	21	10	FF	F3	D2
8	CD	0C	13	EC	5F	97	44	17	C4	A7	7E	3D	64	5D	19	73
9	60	81	4F	DC	22	2A	90	88	46	EE	B8	14	DE	5E	0B	DB
A	E0	32	3A	0A	49	06	24	5C	C2	D3	AC	62	91	95	E4	79
B	E7	C8	37	6D	8D	D5	4E	A9	6C	56	F4	EA	65	7A	AE	08
C	BA	78	25	2E	1C	A6	B4	C6	E8	DD	74	1F	4B	BD	8B	8A
D	70	3E	B5	66	48	03	F6	0E	61	35	57	B9	86	C1	1D	9E
E	E1	F8	98	11	69	D9	8E	94	9B	1E	87	E9	CE	55	28	DF
F	8C	A1	89	0D	BF	E6	42	68	41	99	2D	0F	B0	54	BB	16

CLO 3

Q4: Numerical questions

[2 + (3 + 2 + 2) marks]

- A. In an RSA crypto system, you intercept the ciphertext $C = 10$ sent to a user whose public key is $e = 5$, $n = 35$. What is the plaintext M ?
- B. Alice and Bob are two business partners negotiating a sensitive deal. One of them is connected to an insecure public Wi-Fi network at an airport. To protect their communication, they decide to use the Diffie-Hellman Key Exchange to establish a shared secret key. Once the key is agreed upon, they plan to use it with a symmetric encryption algorithm such as AES to secure all further messages.

They publicly agree on two parameters: Prime $p = 11$, and Generator $g = 6$

Alice selects her private number $a = 7$ and computes her public value A .

Bob selects his private number $b = 8$ and computes his public value B .

Meanwhile, a hacker named Mallory, who is on the same airport Wi-Fi, launches a man-in-the-middle attack. She intercepts and modifies all messages between Alice and Bob. Mallory chooses her private number $m = 4$ and computes her public value M . She then sends M to Bob while pretending to be Alice, and sends the same value M to Alice while pretending to be Bob. As a result:

Alice computes a shared secret K_{AM} with Mallory, believing it to be with Bob.

Bob computes a shared secret K_{BM} with Mallory, believing it to be with Alice.

Neither Alice nor Bob shares the same key with each other — only Mallory knows both.

Questions

- Compute the public values A , B and M , as well as the shared keys K_{AM} and K_{BM} .
- Explain why this situation is catastrophic for Alice and Bob. What can Mallory do at this stage?
- What is the fundamental weakness in the basic Diffie-Hellman protocol that allows this attack?

Information Security (CS3002)

Date: September 21st 2024

Course Instructor(s)

AK, AIS, MZH, SMI, AH, RAR

Sessional-I Exam

Total Time (Hrs): 1

Total Marks: 40

Total Questions: 4

Roll No

Section

Student Signature

Do not write below this line

Note: Students will also write the formula used in each calculation, no direct answer will be accepted. If you think some information is missing then make an assumption and write it clearly. Write all the answers including MCQs on the given answer sheet.

CLO-1	Explain key concepts of information security such as design principles, cryptography, risk management
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CLO #: 1

Question No. 1: Choose the correct options(s).

[5 marks]

1. What is the default access level to an object or resource according to fail-safe defaults?
 - a. Read-only
 - b. Write-only
 - c. **None**
 - d. Full access

2. Which mathematical operation is central to the security of the Diffie-Hellman algorithm?
 - a. Matrix multiplication
 - b. **Modular exponentiation**
 - c. Symmetric encryption
 - d. Factorization of large primes

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3. In the DES algorithm the round key is _____ bit and the Round Input is _____ bits.
- 48, 32
 - 64, 32
 - 56, 24
 - 32, 32
4. Passwords and/or sensitive data are read by (unauthorized) third parties ____
- Sniffing Attack**
 - Spoofing Attack
 - Identity Theft Attack
 - Shadow Server
5. In the RSA algorithm, what is the relationship between the public and private keys?
- Both keys are identical
 - The private key is the inverse of the public key modulo the totient $\phi(n)$**
 - The public key is used for decryption and the private key for encryption.
 - The private key and public key are randomly generated and unrelated.

CLO #: 1

Question No. 2

[5+5+5 marks]

A. Use the Rail Cipher to decrypt the following ciphertext, assuming a depth level of 3:

TPLEATA_NRM_ILRLCT_OKRSOUW_OE_UO (5)

TRUMP WILL RELOCATE TO KASUR

T			P			L			E			A			T			A			-				N						
	R		M		-		I		L		R		L		C		T		-		O		K		S		R		S		O
		U				W			-				O			E			-					U				O			

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B. The following message is encrypted using the Vigenère cipher.

UQCG OQCG UQSV UQCG

What possible length could the keyword have and why? (5)

- 1- Find repeated strings in the ciphertext. Their distance is expected to be a multiple of the key length.
- 2- Compute the gcd of (most) distances.

Diagraph	First Position	Second Position	Distance	Factors
UQ	1	9	8	2,4
UQ	9	13	4	2
CG	3	7	4	2
CG	7	15	8	2,4

Hence, possible key length is 2.

C. Encrypt the following plaintext “**samosa is ready**” using the Row Transposition Cipher, with key “**nick**” (if your roll number ends with an even digit) or key “**back**” (if your roll number ends with an odd digit). (5)

CLO #: 3

Question No. 3:

[10+5 marks]

A. In the RSA algorithm, given that the **p = 61** and **q=53**. You need to calculate the following items:

1. The value of n. 3233
2. The totient $\phi(n)$. 3120
3. If the public exponent $e=17$ private key exponent d. 2753 (2+3+5)

B. Suppose Alice and Bob wish to perform Diffie Hellman key exchange. They agree on prime number $p = 11$ and generator $g = 6$. Alice chooses a secret 5, while Bob chooses 8. An attacker Mallory is listening to all of their communication. He chooses his own secret 4.

Show the calculation that proves that Mallory can perform a man-in-the-middle attack against Alice and Bob’s key exchange. (5)

Alice public = $6^5 \bmod 11 = 10$

Mallory public = $6^4 \bmod 11 = 9$

Bob public = $6^8 \bmod 11 = 4$

Key between Alice and Mallory

Alice calculates $K_1 = 9^5 \bmod 11 = 1$

Mallory calculates $K_1 = 10^4 \bmod 11 = 1$

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Key between Bob and Mallory

Bob calculates $K_2 = 9^8 \bmod 11 = 3$

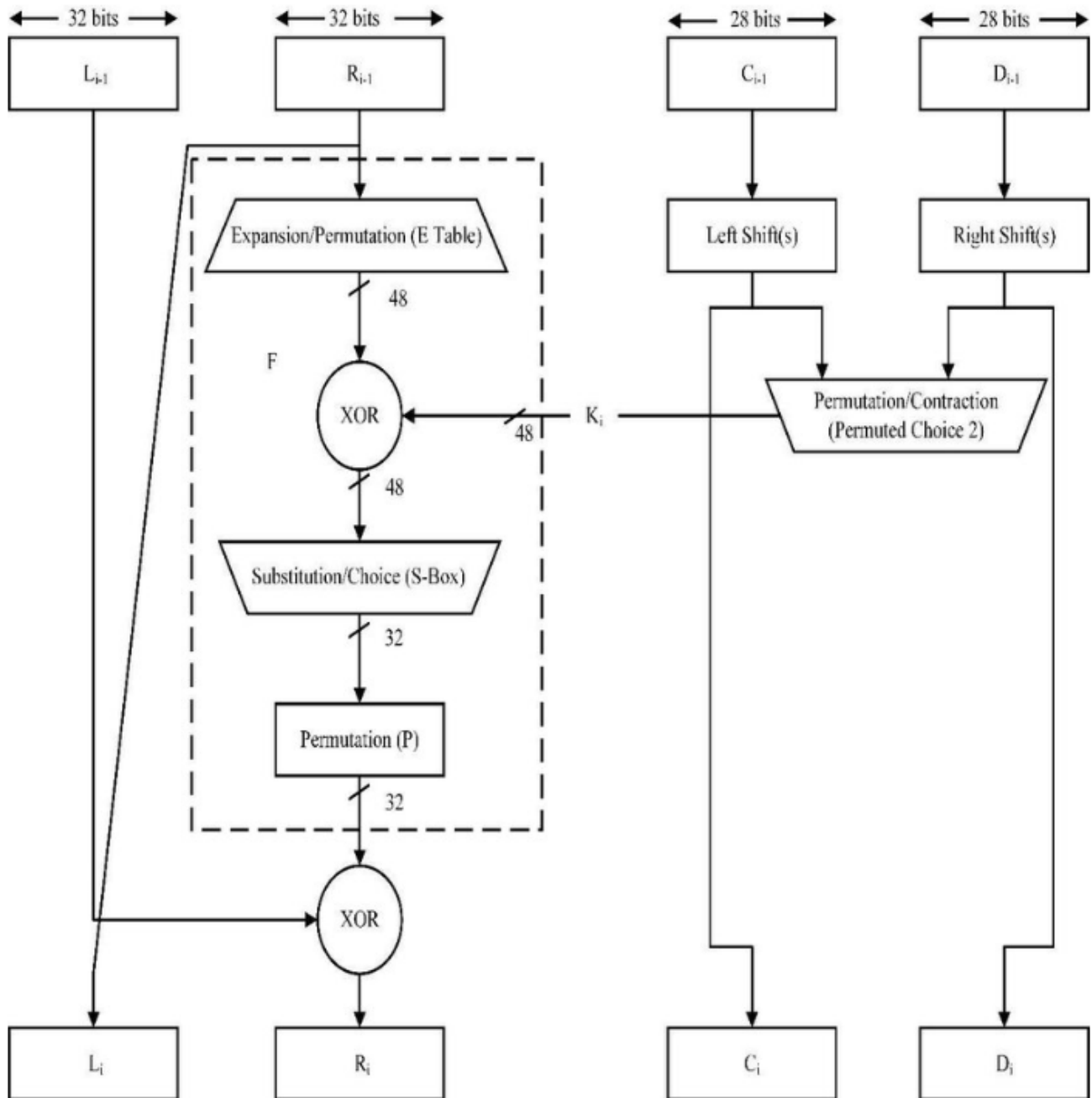
Mallory calculates $K_2 = 4^4 \bmod 11 = 3$

CLO #: 1

Question No. 4:

[5 marks]

Draw the block diagram of a DES single round. Proper label the diagram.



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	Course Name:	Information Security	Course Code:	CS3002
	Program:	BS(Computer Science)	Semester:	Fall 2023
	Duration	60 minutes	Total Marks:	35
	Date:	02-10-23	Weight	12.5
	Exam Type:	Midterm-I	Pages	4

Student : Name: _____ Roll No. _____ Section: _____

Instruction: If you think some information is missing then make an assumption and write it clearly.

Question 1: [CLO:1] [5 marks]

- The process of identifying vulnerabilities and threats and their impact and probability of occurring an attack is called _____.
 - Vulnerability assessment
 - Vulnerability identification
 - Threat detection
 - Risk analysis
- Receiving an SMS message where the sender's number is fake, is an attack against _____.
 - Confidentiality
 - Integrity
 - Availability
 - Authenticity
- Which block cipher mode does not allow parallel encryption of blocks?
 - Cipher Block Chaining (CBC)
 - Counter mode
 - Electronic Code Book (ECB)
 - All of these
- Sam maintains a public key ring in his computer. He will pick a key from the ring during
 - Asymmetric encryption
 - Signing a message
 - Asymmetric decryption
 - Both a and c
- In the DES algorithm the Round Input is 32 bits, which is expanded to 48 bits via _____.
 - Scaling of the existing bits
 - Substitution of the existing bits
 - Addition of zeros
 - Addition of ones

Part I:

Identify the kind of attacks in each of the following statements and identify at least a couple of reasons the people fall for that

- a. You receive a phone call. The caller claims that they are from a bank, and ask your debit card number and OTP for the purpose of account confirmation, otherwise your account will be blocked.

social engineering, or more specifically phishing

people get panic by such threats and make decisions in haste

- b. You tried to login to your flex account during which you ignored a warning displayed by the browser and later on you came to know that your login credentials have been stolen.

phishing attack bcoz it was a look-alike web page operated by attacker

most people do not understand the meaning of SSL warnings, and want to finish their intended task asap.

- c. The outgoing network traffic of your system has suddenly increased and on diagnosis, you have identified that your system is constantly sending a specific request to a server.

This computer has become a zombie, and is participating in a DDoS attack. Attackers infected the computer with a bot malware.

People fall for this attack due to not having good anti malware software, and downloading software from dangerous sources like torrents or piracy sites.

Part II:

We studied following security design principles: Least privilege, Failsafe defaults, Separation of privilege, Economy of mechanism, Psychological acceptability, Complete mediation, Least common mechanism. Analyze the following scenarios and identify, with reasoning, what security design principle is being applied? Also state what advantages are achieved due to that principle.

- A. Marketing staff do not have access to engineering design files.

principle of least privilege. Minimize the chances of access misuse.

- B. Company database is protected by only one layer of encryption rather than two layers.

economy of mechanism, since a single layer is enough. Do not needlessly complicate the system.

Question 3:

[CLO:1]

[2+4+4 marks]

- a. Encrypt the text “voteforme” using Vigenère cipher with a key 245.

Plaintext: v o t e f o r m e
Key: 2 4 5 2 4 5 2 4 5
Ciphertext: x s y g j t t q j

- b. Show the process of Diffie-Helman key exchange using a prime 47 and generator 11. Suppose Alice chooses a secret 9 and Bob chooses 16.

Alice computes $A = 11^9 \bmod 47 = 38$

Bob computes $B = 11^{16} \bmod 47 = 3$ (hard to get it via calculator)

After sharing public keys, Alice computes shared key as $K = 3^9 \bmod 47 = 37$

Bob gets $K = 38^{16} \bmod 47 = 37$

- c. What is a digital envelope? Demonstrate its working with the help of an example! What are the advantages of this mechanism?

Use diagram from slides

Advantages

Gain the speed of symmetric encryption, and convenience of PKC

Question 4:**[CLO:1] [10 marks]**

Describe the key development mechanism using RSA when the values of p and q are given to be 7 and 11 respectively? Although there are multiple options for choosing the encryption value 'e', let's choose 7 that fulfill the criteria and then find 'd' accordingly. You should provide all details and outcome should be written in the form of private and public keys. Demonstrate the working of your system by encrypting a number and then retrieving it back using decryption.

$$p = 7, q = 11$$

$$n = 77$$

$$\phi = 60$$

$$\text{given } e = 7$$

$$\text{Solving } 7 \times d \bmod \phi = 1, \text{ we get smallest } d = 43$$

$$\text{Public key} = 7, 77$$

$$\text{Private key} = 43, 77$$

Encryption of message, say 5

$$5^7 \bmod 77 = 47$$

Decryption

$$47^{43} \bmod 77 = 5$$

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	Course Name:	Information Security	Course Code:	CS3002
	Degree Program:	BS (CS)	Semester:	Fall 2022
	Exam Duration:	60 mins	Total Marks:	35
	Paper Date:	26/09/22	Weight:	12.5
	Exam Type:	Mid I	Page(s):	4

Student : Name: _____ Roll No. _____ Section: _____

Instruction: If you think some information is missing then make assumption and write it clearly.

Question 1: MCQs and True/False

[10 Marks] [CLO 1]

1.1 A way of checking whether the private key matching the public key in a certificate has been compromised and so the certificate should no longer be accepted.

- a) registration list
- b) rejection list
- c) revocation list
- d) all of the above

1.2 A hash function is regarded as strong if a single bit change in the input changes the output by at least ____.

- a) 40%
- b) 50%
- c) 60%
- d) 70%

1.3 When someone takes the place of a legitimate host, it is called ____.

- a) Connection hijacking
- b) IP Spoofing
- c) Phishing
- d) None of the above

1.4 Triple DES algorithm uses a data block size of ____ bits.

- a) 56
- b) 64
- c) 112
- d) 168

1.5 Which cyber security principle states that Security mechanisms should be as simple and small as possible?

- a) Complete mediation
- b) Least Privilege
- c) Economy of mechanism
- d) Open Design

1.6 Applying safeguards that eliminates or reduces residual risk is called defense strategy.

- a) Avoidance
- b) Acceptance
- c) Mitigation
- d) Transference

1.7 Contingency planning consists of

- a) Incidence response planning
- b) Disaster recovery plan
- c) Business continuity plan
- d) All of the above

1.8 Asymmetric cryptography is faster in computation compared to symmetric cryptography.

- a) True
- b) False

1.9 If Alice has a message to send to Bob and she wants to encrypt the message using asymmetric cryptography so that no one other than Bob can read it, she does so by using Bob's public key.

- a) True
- b) False

1.10 Weakness of an asset is called threat

- c) True
- d) False

Question 2:

[6 marks] [CLO 1]

In each of the following scenarios, identify which security design principle is followed OR violated. Some of the security design principle are listed above in MCQ 1.5. Elaborate your response.

- A. The company behind Telegram messenger hired some brilliant cryptographers to design proprietary security protocols for secret chats.

Violation of Open Design because proprietary protocols can not be subject to public scrutiny by experts.

- B. Staff at an airline booking office is required to provide their password as well as scan their smart card before confirming a booking.

Separation of Privileges is being followed, as multiple security conditions are being checked.

- C. A program successfully opens a file in write mode, but after a few hours, writing data to file fails due to permission errors.

Complete Mediation is being followed because initially the program had permissions. After some time the permissions were revoked. The operating system did not rely one-time check, instead it applied security checks on each access.

Alternate answer: Principle of least-privilege is followed because the program was given a temporary privilege escalation, which was taken back after some time.

Question 3:

[3+3+5 marks] [CLO 1]

- a) Suppose you have received the following encrypted messages from the sender and you have been told that RSA algorithm is used to encrypt this information and the public key to decrypt the message is {7, 187}. Your task is to retrieve the original messages.

C1 = 16

C2 = 24

$$16^7 \bmod 187 = 135$$

$$24^7 \bmod 187 = 29$$

- b) What is Euler Totient function (ϕ). What is the ϕ (Φ) of 19?

Φ of a number n counts all co-prime numbers smaller than n .

ϕ (Φ) of 19 = 18 as 19 is a prime number

- c) Analyze the following statements for correctness. Justify with arguments.

- i. Encryption with RSA is not vulnerable to man in the middle attacks.

RSA by itself is vulnerable to MITM attacks because attacker can share their own public key with both parties without either of them realizing.

Alternate Answer: It is not vulnerable as in this mechanism the public key will be certified by the Certification Authority.

- ii. A strong hash function like SHA-512 alone can compute the Message Authentication Code (MAC).

Hash function alone is not sufficient. MAC computation also requires a secret key as input, such as a pre-shared key or the private key of the sender.

Question 4:

[4+4] [CLO 4]

User A wants to download an operating system image from the vendor website. On downloading the OS image, the user should be able to assess the integrity and authenticity of the OS image in an optimal way. Draw a diagram to illustrate the process to ensure and validate integrity and authenticity of the OS image. Briefly describe each step of the process illustrated in your diagram.

Diagram:

Figure on Lecture 5-6 slide # 29 BUT with an important change:

- at the sending side, the key K is the private key of OS vendor
- at receiving side, K is public key of OS vendor

Process Steps:

Vendor will have to create and attach MAC to in order to prove the integrity and authenticity of the downloaded image. Vendor signs the MAC with their private key. User downloads the image and verifies the MAC with vendor's public key.